

Algebra 1
Unit 14
Lesson H2 Practice

Name Answer Key
Date _____
Period _____

High school students in the United States were invited to complete an online survey in 2020. Part of the survey included questions about superhero powers. More than 1,000 students responded to this survey that included a question about a favorite superhero power. Researchers randomly selected 450 of the completed surveys. The two-way table broken down by gender the results.

	To Fly	Freeze Time	Invisibility	Super Strength	Telepathy	Total
Females	10.9%	13.3%	10.7%	0.2%	15.6%	50.7%
Males	11.3%	15.8%	6%	5.6%	10.6%	49.3%
Total	22.2%	29.1%	16.7%	5.8%	26.2%	100%

1. What is the joint relative frequency for females who selected invisibility as their favorite superpower?

10.7%

2. What is the marginal relative frequency for freeze time?

29.1%

3. What is the difference in the joint relative frequencies for males and for females who selected to fly as their favorite superpower?

Males – Females

$$11.3 - 10.9 = 0.4$$

4. Interpret the value of 15.8% in the context of the data in the table.

15.8% of students are males who chose “Freeze Time” as their preferred superpower.

5. Interpret the value of 26.2% in the context of the data in the table.

26.2% of students chose telepathy as their preferred superpower.

6. Interpret the value of $\frac{11.3\%}{100\%}$ in context of the data in the table.

11.3% of the students surveyed were males who wanted to fly.

7. Interpret the value of $\frac{11.3\%}{49.3\%}$ in context of the data in the table.

22.9% of total males wanted to fly.

8. What percentage of females chose super strength as their favorite superpower?

$$\frac{0.2\%}{50.7\%} \approx 0.4\%$$

9. Is there a noticeable difference between the genders and their favorite superpowers?

Not a noticeable difference except with "Super Strength"

10. Is there evidence of an association between gender and a favorite superpower? Explain why or why not.

In general, because the relative frequencies don't all have a noticeable difference, there is not an association.